

Technical Indicators & Strategy Backtesting

Strategy Descriptions

Buy & Hold

The Buy & Hold strategy purchases the selected stock at the beginning of the testing period and holds it until the end without making any additional trades. This strategy serves as the benchmark for comparing the active trading strategies.

Strategy 1 – Trend Following

The Trend Following strategy attempts to profit from sustained price trends. It enters positions only when momentum is positive and the trend is considered strong.

Entry Rule

- MACD is above its signal line.
- ADX is greater than 25.

Exit Rule

- MACD falls below its signal line.
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Strategy 2 – Mean Reversion

The Mean Reversion strategy assumes that prices will eventually return toward their historical average after becoming overextended.

Entry Rule

- RSI is below 35.
- Closing price is below the lower Bollinger Band.

Exit Rule

- RSI is above 65.
- Closing price is above the upper Bollinger Band.

Strategy 3 – Custom Strategy

The Custom Strategy combines indicators from multiple categories (trend, momentum, volatility, and volume) to identify high-quality buying opportunities while limiting downside risk.

Entry Rule

- Closing price is at least four ATR below the 200-day SMA.
- RSI is below 40.
- OBV is above its 20-day moving average.

Exit Rule

- Closing price is at least five ATR above the 200-day SMA.
 - RSI is above 70.
 - MACD crosses below its signal line.
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Performance Comparison

Strategy	Total Return	CAGR	Volatility	Sharpe	Sortino	Max Drawdown	Win Rate	Trades
Trend Following	-47.60%	-11.73 %	16.57%	-0.66	-0.51	-53.64%	30.43%	46
Mean Reversion	13.40%	2.46%	21.17%	0.22	0.22	-37.45%	77.78%	9
Custom Strategy	52.63%	8.63%	16.04%	0.60	0.49	-26.99%	100.00 %	2
Buy & Hold	47.47%	7.79%	26.73%	0.42	0.61	-37.56%	N/A	0

Discussion of Results

The four strategies produced noticeably different results over the testing period. The Trend Following strategy performed the worst, producing a negative total return and the largest

maximum drawdown. This suggests that the selected trend signals were not well suited to the market conditions during the backtest.

The Mean Reversion strategy generated a positive return with a relatively high win rate, although it executed only a small number of trades. While its overall return was modest, it demonstrated that buying oversold conditions can be effective during certain market environments.

The Custom Strategy achieved the strongest overall performance. It produced the highest total return, the highest CAGR, the highest Sharpe Ratio, and the smallest maximum drawdown among all strategies. By combining trend, momentum, volatility, and volume indicators, it was able to identify high-probability entry points while avoiding many of the large drawdowns experienced by the other strategies. Although it completed only two trades during the testing period, both were profitable, resulting in a 100% win rate.

Buy & Hold remained a strong benchmark, benefiting from Microsoft's long-term price appreciation. However, it also experienced substantially higher volatility and a larger drawdown than the Custom Strategy. This demonstrates that maximizing return alone does not necessarily produce the best investment strategy. Risk-adjusted measures such as the Sharpe Ratio and Maximum Drawdown indicate that the Custom Strategy delivered a more favorable balance between return and risk for this dataset.